

SIMATS ENGINEERING

ASSESSMENT TOOL 3 – TRAVERSAL CHALLENGE

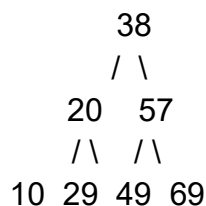
UNIT 3: Binary Tree, Binary Search Tree and AVL Tree

Course Name: Data Structures **Course Code:** CSA0310

Priya dharshini M (192521051)

CHALLENGE 17 – QUESTIONS AND COMPLETE ANSWERS

A. Binary Tree (4 marks): Write the preorder, inorder, postorder and level-order traversals of the given tree.



Answer – Preorder: Root → Left → Right

Visiting order: 38 → 20 → 10 → 29 → 57 → 49 → 69

Answer – Inorder: Left → Root → Right

Visiting order: 10 → 20 → 29 → 38 → 49 → 57 → 69

Answer – Postorder: Left → Right → Root

Visiting order: 10 → 29 → 20 → 49 → 69 → 57 → 38

Answer – Level-order: Level by level, left to right

Level 1: 38

Level 2: 20 → 57

Level 3: 10 → 29 → 49 → 69

Visiting order: 38 → 20 → 57 → 10 → 29 → 49 → 69

B. BST (3 marks): Insert the keys in this order: 38, 20, 57, 12, 28, 50, 69. Draw the final BST and write its postorder traversal.

Solution – Insertion Steps

Step 1 – Insert 38

Step 2 – Insert 20 ($20 < 38$, so go left)

```
  38
 /
20
```

Step 3 – Insert 57 ($57 > 38$, so go right)

```
  38
 /  \
20   57
```

Step 4 – Insert 12 ($12 < 38$ and $12 < 20$, so go left of 20)

```
  38
 /  \
20   57
 /
12
```

Step 5 – Insert 28 ($28 < 38$ but $28 > 20$, so go right of 20)

```
  38
 /  \
20   57
 /  \
12   28
```

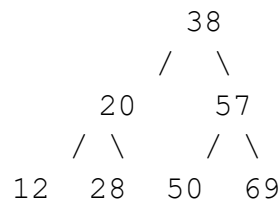
Step 6 – Insert 50 ($50 > 38$ but $50 < 57$, so go left of 57)

```
  38
 /  \
20   57
 /  \  /
12  28 50
```

Step 7 – Insert 69 ($69 > 38$ and $69 > 57$, so go right of 57)

```
  38
 /  \
20   57
 /  \  /  \
12  28 50  69
```

Final BST:



Postorder Traversal: Left → Right → Root

Visiting order: 12 → 28 → 20 → 50 → 69 → 57 → 38

Final Answer: 12 → 28 → 20 → 50 → 69 → 57 → 38

C. AVL Tree (3 marks): Insert 75, 50, 25 into an empty AVL tree. Name the rotation, draw the balanced tree and write its preorder traversal.

Solution – Insertion Steps

Step 1 – Insert 75

```
75
```

Step 2 – Insert 50 (50 < 75, so insert to the left)

```
75
 /
50
```

Step 3 – Insert 25 (25 < 75 and 25 < 50, so insert to the left of 50)

```
75
 /
50
 /
25
```

Balance condition: Node 75 becomes unbalanced because the insertion is in the left subtree of its left child.

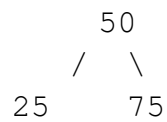
Imbalance: LL (Left-Left)

Rotation: Right rotation at node 75

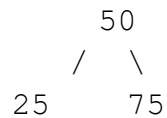
Before Rotation:

```
75
 /
50
 /
25
```

After Right Rotation:



Final Balanced AVL Tree:



Preorder Traversal: Root → Left → Right

Visiting order: 50 → 25 → 75

Final Answer: 50 → 25 → 75

FINAL ANSWER SUMMARY

Part	Answer
A – Preorder	38 → 20 → 10 → 29 → 57 → 49 → 69
A – Inorder	10 → 20 → 29 → 38 → 49 → 57 → 69
A – Postorder	10 → 29 → 20 → 49 → 69 → 57 → 38
A – Level-order	38 → 20 → 57 → 10 → 29 → 49 → 69
B – BST Postorder	12 → 28 → 20 → 50 → 69 → 57 → 38
C – AVL Rotation	LL imbalance → Right rotation at 75
C – AVL Preorder	50 → 25 → 75